Freight GHG emissions report

Organisation	Example Freight A/S
Reporting period	1 Jan 2026 to 31 Mar 2026
Level	Transport or hub services (GLEC Framework v3.3, Section 2, Chapter 1)
GHG Protocol	Scope 3, category 4 (upstream transportation and distribution)
Emission factors	desnz-2026+nl-iww-2026+eu-mrv-2024+glec-3.3
Calculated	2 November 2026, 22:49 UTC

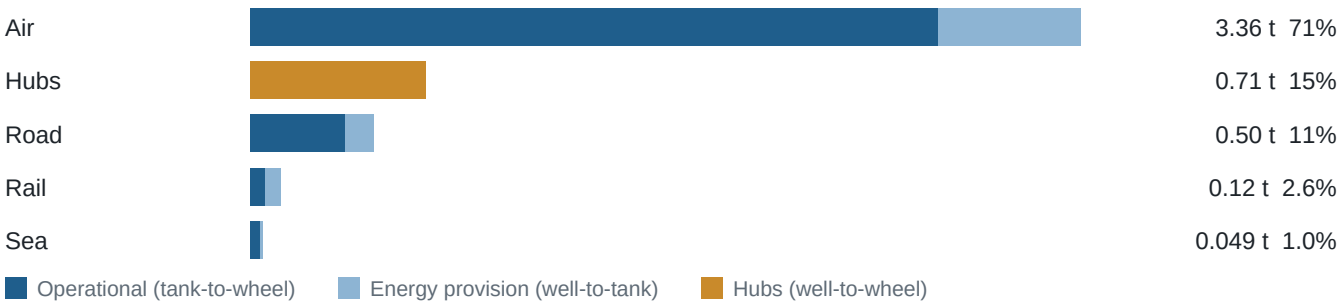
Calculated with Sootcase following the GLEC Framework v3.3 and ISO 14083:2023 methodology. Conformity with ISO 14083 is not declared by this report.

Key figures

Total emissions (WTW)	Transport, operational (TTW)	Transport, energy provision (WTT)	Hubs (WTW)
4.74 t CO2e	**3.27** t CO2e	**0.76** t CO2e	**0.71** t CO2e

Transport activity	Transport intensity (WTW)	Shipments calculated	Cargo
19,683 tonne-km	**204.7** g CO2e / tonne-km	**3 of 4** 1 failed	**35.5** tonnes

Emissions by mode of transport



Mode	Legs	Tonne-km	TTW kg CO2e	WTT kg CO2e	WTW kg CO2e	g CO2e/tkm	Share of transport
Air	1	2,379	2,783	578.1	3,361	1,413.0	83%
Road	1	3,984	382.5	115.5	498.0	125.0	12%
Rail	1	6,600	59.2	62.0	121.2	18.4	3.0%
Sea	1	6,720	40.6	8.74	49.3	7.3	1.2%
Transport	**4**	**19,683**	**3,265**	**764.3**	**4,030**	**204.7**	**100%**

Emissions at hubs

Hub type	Stops	Tonnes handled	WTW kg CO2e	g CO2e/t
Warehouse	1	20.0	710.0	35,500.0
All hubs	**1**		**710.0**	

Hub factors are well-to-wheel only, so hubs are not split into operational and energy provision emissions.

Emissions by energy carrier

Energy carrier	TTW kg CO2e	WTT kg CO2e	WTW kg CO2e	Share of transport
Unspecified	3,265	764.3	4,030	100%

"Unspecified" is transport calculated from default or own intensity values, which don't name the fuel or electricity used.

GHG Protocol scopes

For the company buying the transport

Scope	kg CO2e
Scope 3, category 4: upstream transportation and distribution	4,740

For the operator of the vehicles and hubs

Scope	kg CO2e
Scope 1: fuel burnt in the vehicles	3,265
Scope 2: electricity used by the vehicles	0
Scope 3, category 3: producing and delivering the fuel and energy	764.3
Hubs (well-to-wheel, not split by scope)	710.0

Distances

Share of tonne-km by the type of distance used (ISO 14083 asks for it to be stated).

	Shortest feasible	Great circle	Actual	Not stated
Air		100%		
Road				100%
Rail				100%
Sea				100%
All transport		**12%**		**88%**

Data quality

Primary data 0.0% Modelled 0.0% Default values 100%

Shares of total well-to-wheel emissions. Primary: measured fuel or the operator's own values. Modelled: from shipment inputs such as load or energy use. Default: published emission intensities.

Sources of default values

Source	WTW kg CO2e	Share
GLEC Framework v3.3 (Smart Freight Centre, 2026), Section 3 Module 2, Table 1, air: aircraft type unknown, short-haul (< 1500 km), including the +95 km DAF, for use with the great circle distance	3,361	71%
GLEC Framework v3.3 (Smart Freight Centre, 2026), Section 3 Module 2, Table 3, logistics hubs (GILA project: Fraunhofer IML, Politecnico di Milano, GreenRouter; median values): warehouse, ambient (n = 92)	710.0	15%
GLEC Framework v3.3 (Smart Freight Centre, 2026), Section 3 Module 2, Europe and South America road, Table 8: rigid truck 26-32 t GVW, average/mixed load, diesel; includes GLEC's +5% distance conversion, so it is per shortest feasible distance	498.0	11%
GLEC Framework v3.3 (Smart Freight Centre, 2026), Section 3 Module 2, rail, Europe: EU average where the traction is unknown, 18.4 g CO2e/t-km WTW (p. 98, based on 62% of EU track being electrified); the TTW/WTT split weights GLEC's Table 4 diesel and Table 5 electric average/mixed values 38% / 62%	121.2	2.6%
GLEC Framework v3.3 (Smart Freight Centre, 2026), Section 3 Module 2, Tables 18-19, container vessels (Clean Cargo trade lane factors, reporting year 2025): industry average (origin-destination pair unknown), dry, 13 + 60.4 g CO2e/TEU-km (WTT + TTW) / 10 t per TEU (GLEC's average when the cargo mass is not known); includes GLEC's +15% distance conversion, so it is per shortest feasible distance	49.3	1.0%

Largest shipments (3 of 3)

Shipment	Route	Tonnes	Tonne-km	WTW kg CO2e	g/tkm	Share
A2	Copenhagen Kastrup Airport (CPH) -> Frankfurt Main Airport (FRA) (air)	3.50	2,379	3,361	1,413.0	71%
A3	rail	20.00	6,600	831.2	18.4	18%
A1	road, sea	12.00	10,704	547.3	51.1	12%

Coverage

Shipments in the input	4
Calculated and included	3
Failed and excluded	1
Transport legs	4
Hub stops	1
Cargo, tonnes	35.5

Why shipments failed

Reason	Shipments
line #: weight_kg: "abc" is not a number	1

Failed shipments (1 of 1)

Shipment	Error
BAD	line 7: weight_kg: "abc" is not a number

Notes on omissions and deviations

- 1 of 4 shipments could not be calculated and are excluded from all totals (BAD). Reasons: 1 × line #: weight_kg: "abc" is not a number.
- Hub emissions are reported well-to-wheel only: the hub factors used carry no split into operational and energy provision emissions. Operational (TTW) and energy provision (WTT) totals therefore cover transport only.
- For 87.9% of tonne-km the type of distance was not stated (distances given without distance_type).
- Tonne-km combine more than one type of distance; see the shares per mode.
- Default emission intensities are used where no primary data (measured fuel or operation category values) was supplied for a leg or hub; their sources are listed.

Method and supporting information

Per-shipment results with factors, sources and distances: Sootcase batch output for the same input. Method: https://github.com/majidkabir/sootcase

Emissions are calculated per leg as tonne-km × emission intensity (× a distance adjustment factor where the distance type differs from the factor's), or from measured fuel or energy where given, following the GLEC Framework v3.3 and ISO 14083:2023. Every leg's distance, method, data type and factor source are in the per-shipment results. GLEC Framework default values: Smart Freight Centre, used with permission; no affiliation or endorsement is implied.

These are calculations for your own reporting. Sootcase is not certified against ISO 14083; conformity depends on how the data was collected and the boundaries chosen.